Barn 4.0

Continuous monitoring of barn conditions

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Translation intended to support teaching module on the subject.

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Introduction



Introduction

Data collection enabled by the fourth wave of the Internet and the Internet of Things (IoT) approach provides relatively inexpensive methods for continuous monitoring of various operating environments. Understanding the different stages of the food chain is important for improving food safety, promoting the efficient use of raw materials, taking into account consumers' ethical concerns, and mitigating climate change.

In February 2019, environmental monitoring devices measuring temperature, humidity, illuminance, carbon dioxide concentration, and movement (ERSCO2, ELSYS) were installed in the Mustiala research dairy barn located in Tammela and operated by Häme University of Applied Sciences. Our objective is to increase understanding of how the barn's structural characteristics influence the environmental conditions within the facility, and how those conditions in turn affect cow welfare, milking behavior, and milk production.

The selection of measurement devices was originally driven by practical applicability. The aim was to utilize affordable commercial technology that can be easily replicated and that provides meaningful environmental measurements for both farmers and research purposes. The devices operate on the low-power LoRaWAN™ communication network, making long-term measurements possible without a separate power supply. Three devices were installed on support structures located between the cubicles at a height of approximately 1.6 meters. This placement was chosen to position the sensors close to the cows while keeping them out of the animals' reach. A fourth device was installed near the milking robot. In addition, open weather data provided by the Finnish Meteorological Institute were incorporated into the analysis.

Introduction

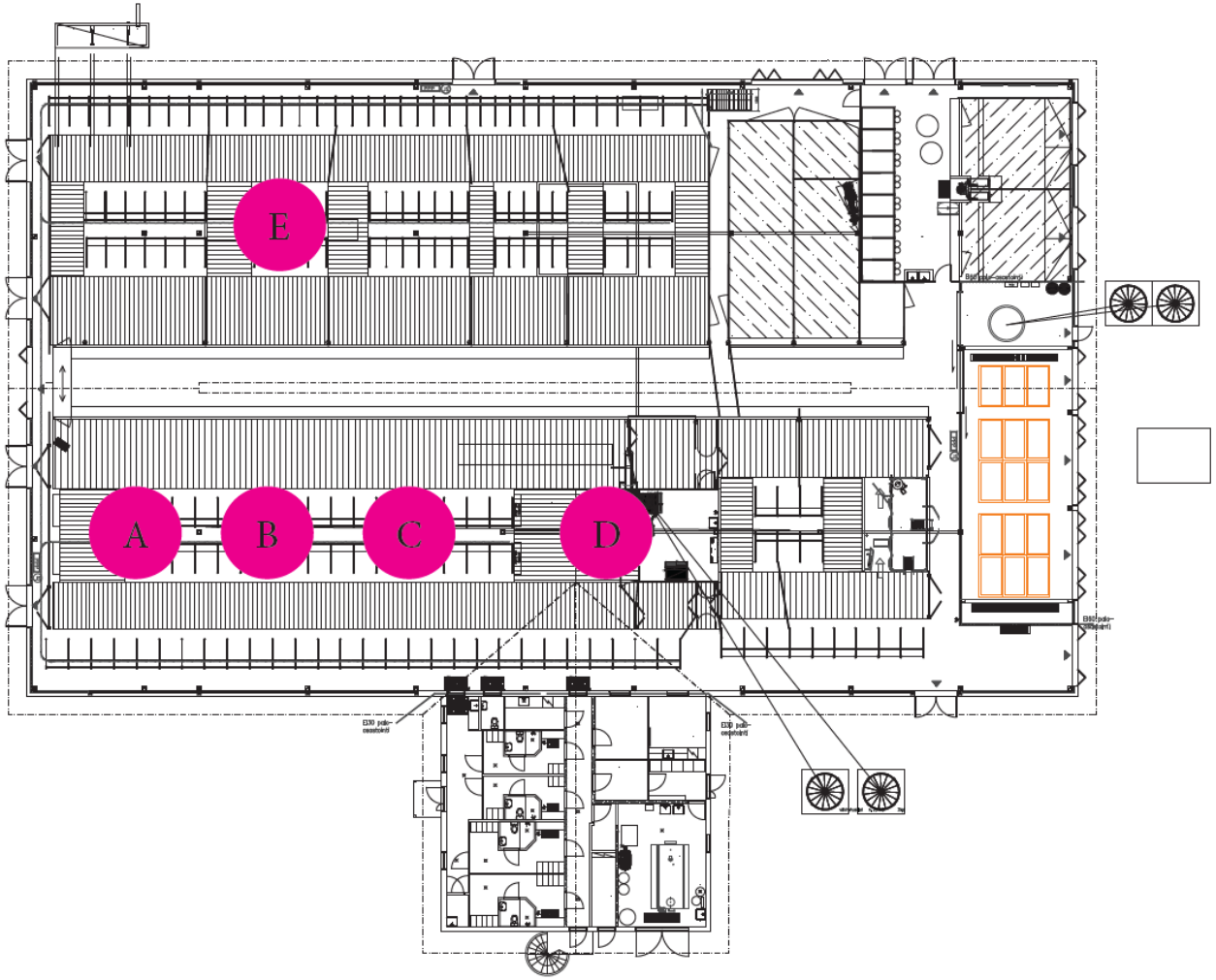
The devices were configured to transmit measurements every 15 minutes. Of these transmissions, 76-80% were successfully received, except for device C, for which the reception rate was 62%. The longest period without any received transmissions lasted between 28 and 33 hours. Two such outages occurred during the monitoring period: one in May and another at the end of June. On average, measurements were received at intervals of 17-19 minutes. The temperature sensor of device C failed on June 17, and the temperature sensor of device B failed on July 10.

Later in 2019, the experimental setup was enhanced with video recording and more detailed planning regarding the placement of measurement points.

Conclusion: Continuous environmental monitoring is straightforward to implement, and data transmission proved to be reliable. However, devices designed for indoor use do not withstand the demanding conditions of a dairy barn environment. During the test period from March 1 to July 31, changes in environmental conditions and milk production appeared to be temporally associated with one another.

Measurement setup

Four ELSYS ERS-CO2 devices (A-D) were installed in February 2019. Most of these devices failed during the summer. In October 2019, replacement ELSYS ELT-2 devices, designed for outdoor use, were installed at the same locations. At measurement point E, two devices were mounted vertically above one another to investigate the effect of sensor installation height on the recorded measurements.



Measurement setup

Devices A, B, and C were installed on support pillars located between the cow stalls, while device D was installed adjacent to the milking robot.





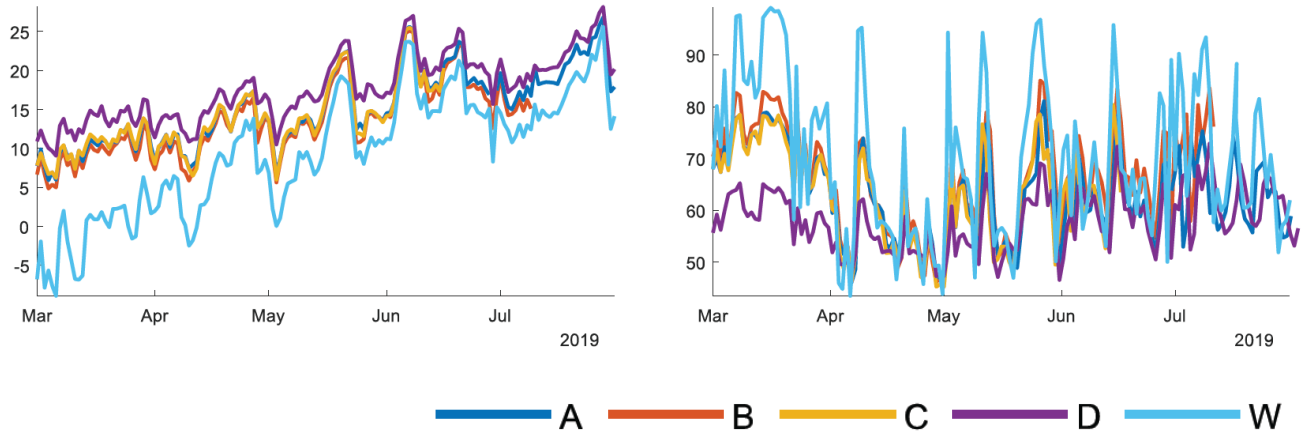
Weather data

The nearest weather station operated by the Finnish Meteorological Institute (FMI) is located in Ilmala, Jokioinen, approximately 14.5 km from the research barn.



Results

Temperature and relative humidity measurements from devices A-D and the weather station (W). Environmental conditions within the cubicle area were relatively homogeneous. The air surrounding device D, located near the milking robot, was consistently warmer and drier than in the cubicle area.



Results

Temperature-Humidity Index (THI) calculated from temperature (T) and relative humidity (RH) measurements collected by devices A-D and from weather station data (W).

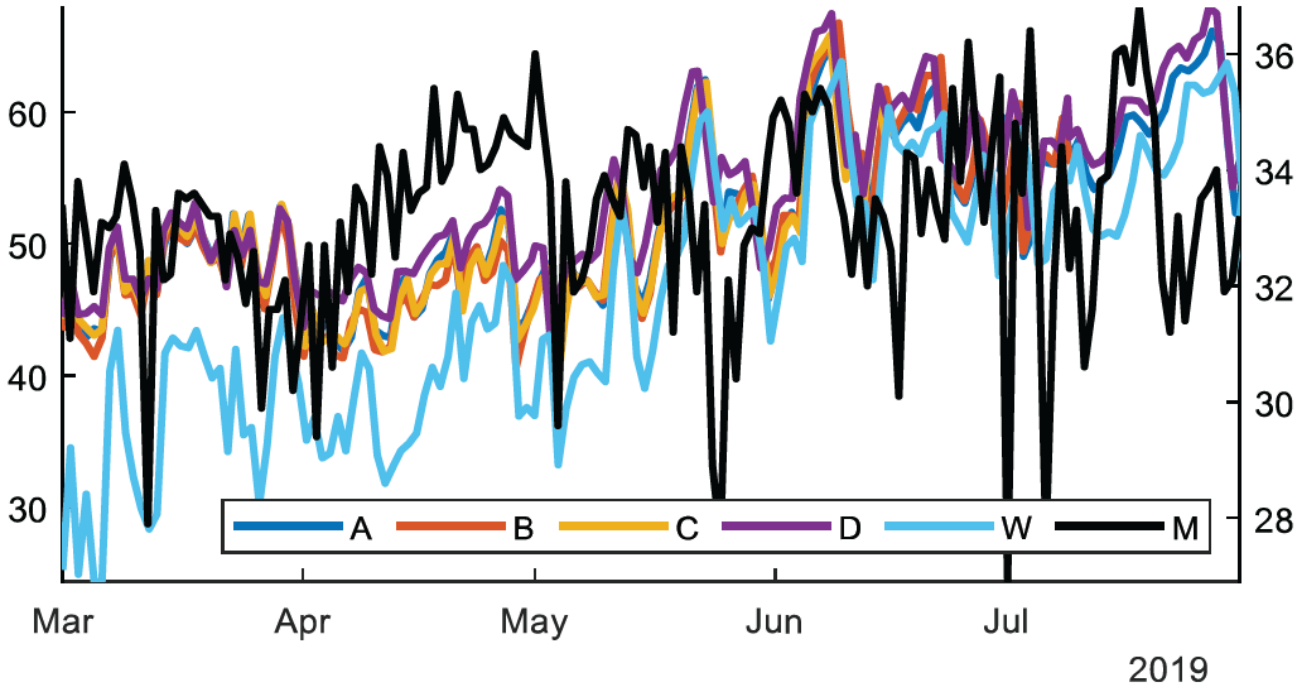
The THI was calculated according to Dikmen & Hansen (2009)

$$THI=Tdb+0.36\times Tdp+41.2$$

where Tdp is the dew point temperature estimated from air temperature (Tdb) and relative humidity. The outdoor and barn environmental conditions differed considerably during March and April.

The trends of the Temperature-Humidity Index and the average daily milk yield (M) closely followed each other throughout the observation period.

Dikmen, S., & Hansen, P. J. (2009). Journal of Dairy Science, 92(1), 109-116.



Results

Continuous monitoring enables the analysis of hour-to-hour or day-to-day variations in environmental conditions and animal performance.

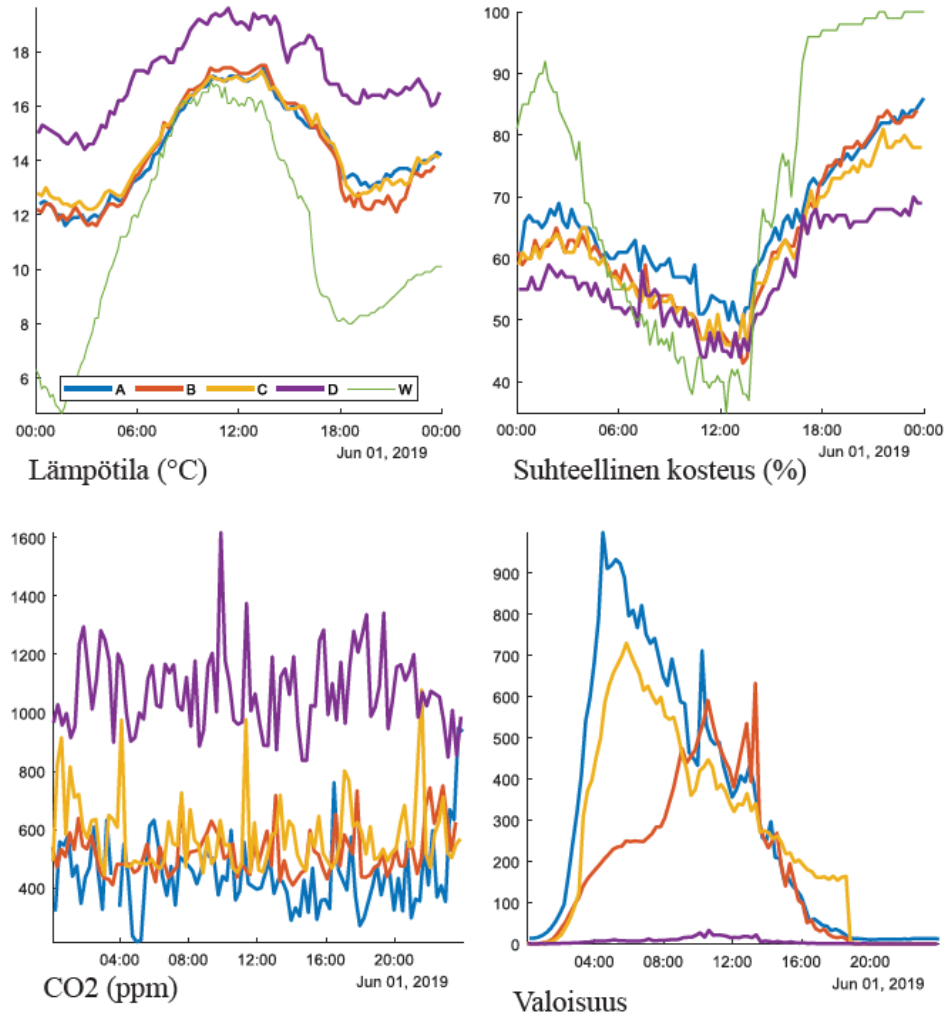
Translations

Lämpötila: Temperature

Suhteellinen kosteus: Relative humidity

CO2: Carbondioxide

Valoisuus: Lighting

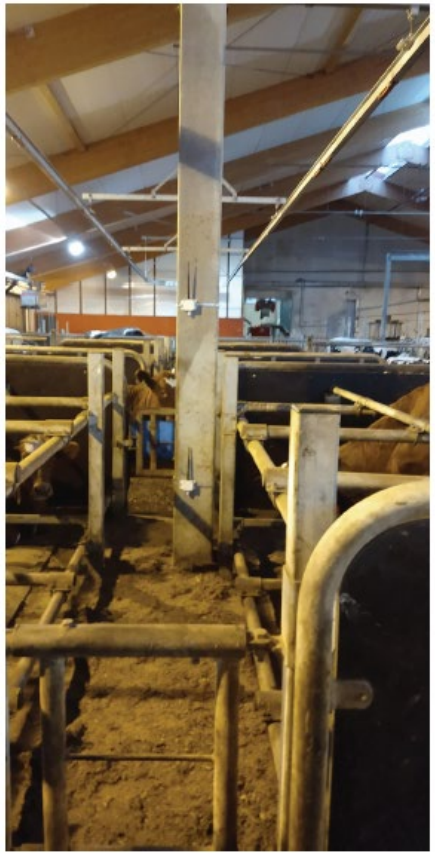


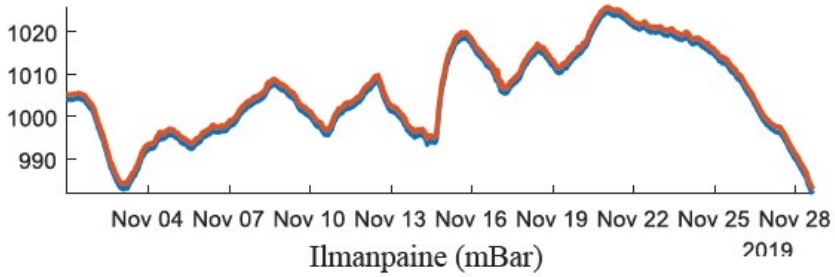
Results

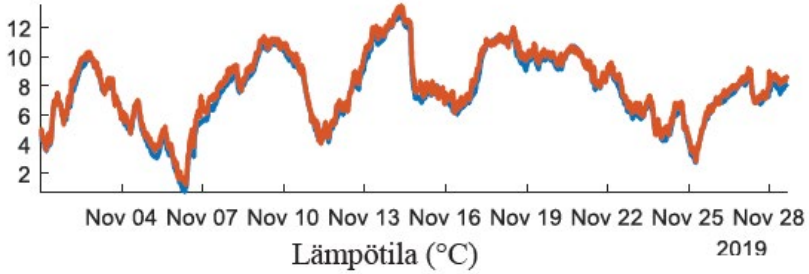
Minimal Effect of Sensor Installation Height on Measurements

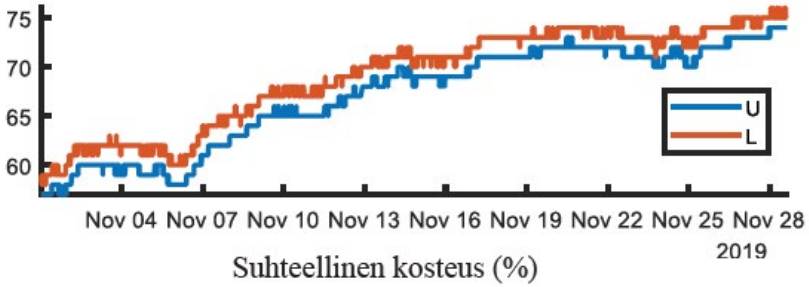
The failed sensors were replaced with IP67-rated ELSYS ELT-2 devices, which measure air temperature, relative humidity, and atmospheric pressure. These replacement sensors were also used to investigate the effect of sensor installation height on measurement results.In a preliminary four-week test, measurements recorded by sensors installed at heights of 50 cm (L) and 160 cm (U) were found to be very similar, indicating that sensor height had only a minor effect on the measured environmental conditions.

Translations. Ilmanpaine: Athmospheric pressure, Lämpötila: Temperature, Suhteellinen kosteus: Relative humidity.









Video-Based Analysis of Cow Activity and Behaviour

Video footage from the barn's surveillance cameras could potentially be used to identify the number of cows, their activities, and their interactions with one another. The objective is to derive information about cow behaviour and welfare from these observations.

By combining computer vision techniques with environmental sensor data, it becomes possible to investigate how factors such as temperature, humidity, and the Temperature-Humidity Index (THI) influence animal movement patterns, occupancy of different barn areas, social interactions, and overall behavioural responses.



Acknowledgements

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